

ILHEUS, Brazil

WMO: 833490

Lat: 14.816S

Lon: 39.033W

Elev: 15

StdP: 14.69

Time Zone: -3.00 (W03)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	64.9	66.2	60.8	79.6	70.4	62.4	84.4	71.2	20.3	75.5	18.5	75.9	7.3	270	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	87.9	79.0	87.1	78.5	86.2	77.9	80.1	85.3	79.7	84.9	79.1	84.3	9.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	149.9	83.7	77.9	145.3	83.0	77.3	142.1	82.5	43.9	85.3	43.2	85.3	42.6	84.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.8	13.4	62.4	88.5	1.7	1.7	61.2	89.7	60.1	90.7	59.2	91.6	57.9	92.8		
(5)			61.0	81.7	1.6	1.8	59.9	83.0	58.9	84.0	58.0	85.0	56.8	86.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	77.4	80.0	80.2	80.2	79.2	77.1	75.2	73.7	74.9	77.0	78.3	79.4	(6)	
(7)		DBStd	3.10	1.71	1.71	1.76	1.86	1.94	1.74	1.87	1.80	2.30	2.46	2.39	1.76	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	9999	929	844	937	877	840	756	736	734	748	838	848	912	(10)
(11)		CDD65	4524	464	424	472	427	375	306	271	269	298	373	398	447	(11)
(12)		CDH74	36930	4815	4420	4850	3760	2535	1596	1178	1258	1838	2884	3458	4338	(12)
(13)		CDH80	8147	1334	1335	1504	1017	465	158	62	53	152	397	647	1023	(13)
(14)	Wind	WSAvg	7.4	7.3	7.0	6.8	7.1	7.5	7.4	7.7	8.0	8.1	8.1	7.4	6.9	(14)
(15)	Precipitation	PrecAvg	63.6	5.4	5.8	6.8	6.0	4.7	5.7	5.8	4.2	3.7	4.6	6.2	4.7	(15)
(16)		PrecMax	99.6	14.4	13.5	16.1	13.9	14.9	19.2	13.3	8.9	8.3	10.6	12.7	13.9	(16)
(17)		PrecMin	38.8	1.5	2.2	2.1	0.4	0.1	0.5	2.6	1.1	1.1	1.0	0.4	0.5	(17)
(18)		PrecStd	15.0	3.0	3.0	3.6	3.3	3.4	3.7	2.4	2.0	1.7	2.2	3.1	2.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	88.2	89.2	88.1	86.2	84.4	82.8	82.6	84.3	85.8	86.4	87.9	(19)
(20)			MCWB	78.8	79.5	79.4	79.5	78.0	75.8	74.5	73.9	75.8	77.3	77.7	78.0	(20)
(21)		2%	DB	86.4	87.6	87.9	86.9	84.6	82.7	81.3	81.0	82.7	84.3	85.6	86.3	(21)
(22)			MCWB	77.8	79.1	79.1	78.7	76.7	74.7	73.3	72.9	74.8	76.3	77.3	77.8	(22)
(23)		5%	DB	86.0	86.3	86.4	86.0	83.9	82.0	80.6	80.5	82.0	82.8	84.3	85.2	(23)
(24)			MCWB	77.6	78.0	78.2	78.1	76.2	74.3	72.9	72.6	74.3	75.3	76.5	77.3	(24)
(25)		10%	DB	85.1	85.7	85.9	84.6	82.6	80.6	79.2	79.2	80.7	82.3	83.3	84.3	(25)
(26)			MCWB	77.1	77.6	77.9	77.3	75.3	73.6	72.3	72.2	73.3	75.0	76.1	76.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.5	80.9	81.0	80.7	79.5	77.4	76.9	75.9	77.1	78.8	79.2	80.0	(27)
(28)			MCDWB	85.1	86.1	86.0	86.5	84.1	82.0	81.0	79.9	82.4	82.9	83.7	84.1	(28)
(29)		2%	WB	79.6	79.9	80.1	79.7	78.3	76.3	75.1	74.7	76.0	77.6	78.3	79.1	(29)
(30)			MCDWB	84.5	85.4	85.4	85.3	83.1	81.0	79.5	79.4	81.3	82.2	82.9	83.5	(30)
(31)		5%	WB	78.8	79.2	79.5	79.0	77.5	75.4	74.1	73.8	75.2	76.9	77.7	78.4	(31)
(32)			MCDWB	84.0	84.7	84.8	84.3	82.2	80.2	78.8	78.8	80.4	81.6	82.4	83.0	(32)
(33)		10%	WB	78.1	78.5	78.9	78.2	76.5	74.6	73.2	72.9	74.4	76.1	77.0	77.8	(33)
(34)			MCDWB	83.4	84.0	84.1	83.4	81.0	79.2	78.0	77.9	79.5	80.9	81.9	82.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	10.7	10.8	10.7	10.8	11.1	11.2	11.4	10.8	9.6	9.3	9.6	(35)
(36)			MCDBR	10.9	11.3	11.5	11.7	11.8	12.3	12.2	12.3	11.3	9.9	10.0	10.6	(36)
(37)			MCWBR	4.6	4.7	4.9	5.3	5.6	6.0	6.0	6.0	5.5	4.7	4.7	4.5	(37)
(38)		5% WB	MCDWB	10.0	10.6	10.8	10.9	10.8	11.1	11.2	11.4	10.5	8.9	9.2	9.3	(38)
(39)			MCWBR	4.8	5.1	5.3	5.5	5.8	6.2	6.3	6.2	5.7	4.8	4.8	4.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.394	0.389	0.390	0.387	0.381	0.369	0.361	0.371	0.423	0.427	0.421	0.412	(40)	
(41)		taud	2.503	2.524	2.536	2.545	2.552	2.572	2.579	2.524	2.367	2.381	2.422	2.445	(41)	
(42)		Ebn at Noon	301	300	293	284	275	273	279	285	279	285	291	296	(42)	
(43)		Edh at Noon	36	36	34	32	30	28	29	32	40	41	39	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2033	1978	1813	1605	1374	1270	1347	1549	1695	1820	1817	1951	(44)	
(45)		RadStd	151	145	128	126	63	65	52	72	119	176	157	164	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page